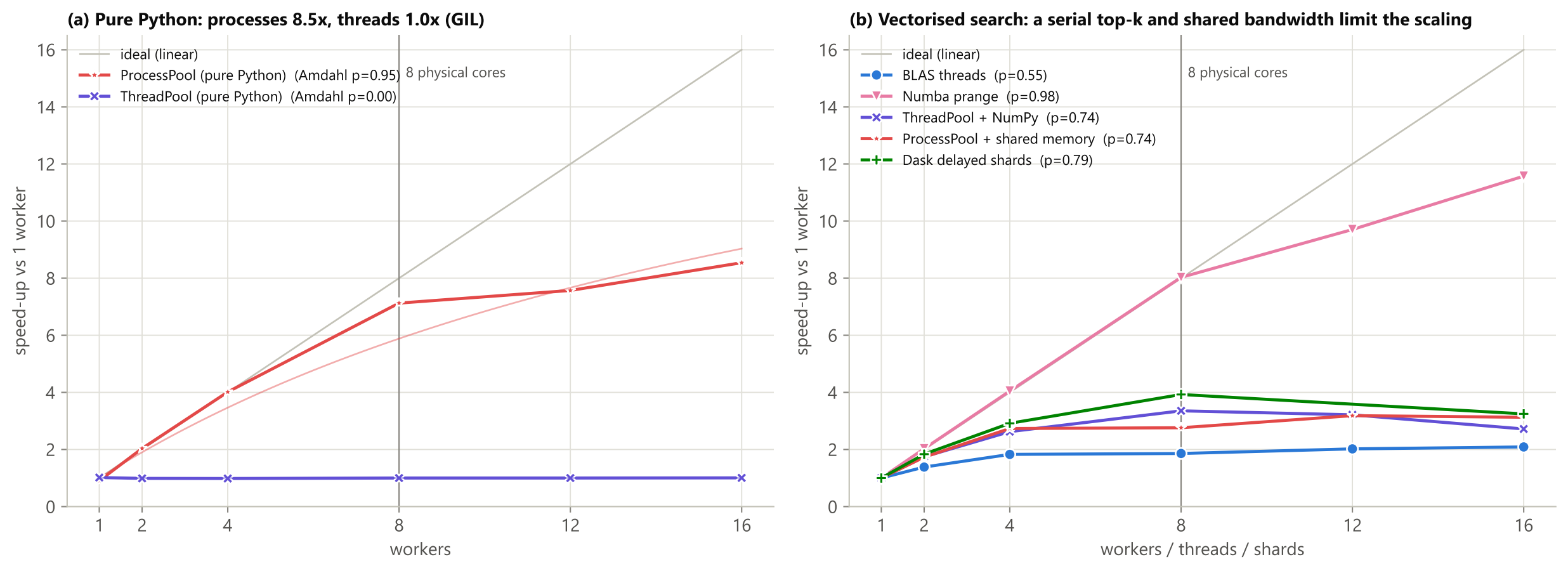
(a) Pure Python: processes 8.5x, threads 1.0x (GIL)
ideal (linear)
ProcessPool (pure Python) (Amdahl p=0.95)
ThreadPool (pure Python) (Amdahl p=0.00)
8 physical cores
speed-up vs 1 worker
workers
(b) Vectorised search: a serial top-k and shared bandwidth limit the scaling
ideal (linear)
BLAS threads (p=0.55)
Numba prange (p=0.98)
ThreadPool + NumPy (p=0.74)
ProcessPool + shared memory (p=0.74)
Dask delayed shards (p=0.79)
8 physical cores
speed-up vs 1 worker
workers / threads / shards